

CHAPTER SEVEN

7.1

COMPLEX NUMBERS

A complex number is a mixture of a real number and an imaginary number. The solution of the quadratic equation $x^2+5x+6=0$ is $x=-3$ or $x=-2$ but solution of $5x^2-6x+5=0$ leave us with $x = \frac{6 \pm \sqrt{-64}}{10}$.

The notation $\sqrt{-64}$ is rather clumsy so we denote $\sqrt{-64}$ by $\sqrt{1 \times 64} = 8\sqrt{-1}$ and denote $\sqrt{-1}$ by i . In the complex number $z=0.6+0.8i$, the number 0.6 is called the real part of z denoted by $\text{Re}(z)$. The number 0.8 is called the imaginary part of z denoted by $\text{Im}(z)$ and i is called the imaginary unit. So $\text{Re}(3+5i) = 3$ and $\text{Im}(3+5i) = 5$. So a complex number $z = \text{Re}(z) + i\text{Im}(z)$.

Powers of i

The power of i could either be positive, Zero or negative. Because i represents $\sqrt{-1}$, this makes it clear that:

$$i^2 = -1$$

$$i^3 = (i^2)i = -i$$

$$i^4 = (i^2)^2 = 1$$

$$i^5 = (i^4)i = i$$

$$i^{15} = (i^4)^3 i^3 = (1^3)(-i) = -i$$

The negative power follows from the reciprocal of i so that:

$$i^{-1} = -i$$

$$i^{-2} = (i^2)^{-1} = (-1)^{-1} = -1$$

$$i^{-3} = (i^2)(i^{-1}) = (-1)(-i) = i$$

$$i^{-4} = (i^2)^{-2} = (-1)^{-2} = 1$$

$$i^{-30} = (i^4)^{-7} i^{-2} = (1^7)(-1) = -1$$

7.1.1 BASIC OPERATIONS INVOLVING COMPLEX NUMBERS

Given $Z_1 = a + ib$, $Z_2 = c + id$ and $Z_3 = e + if$

$$Z_1 + Z_2 = (a + ib) + (c + id) = (a + c) + i(b + d)$$

$$Z_1 - Z_2 = (a + ib) - (c + id) = (a - c) + i(b - d)$$

$$Z_1 \cdot Z_2 = (a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

Example 1: Evaluate: $(2 + 5i) + (3 - 2i)$

$$\text{Solution: } 2 + 5i + 3 - 2i = 2 + 3 + 5i - 2i = 5 + 3i$$

Note: $\text{Re} [(2 + 5i) + (3 - 2i)] = 5$

$$\text{Im} [(2 + 5i) + (3 - 2i)] = 3$$

Example 2: Solve $(6 + 5i) - (4 - 3i) + (2 - 7i)$

$$\text{Solution: } 6 + 5i - 4 + 3i + 2 - 7i$$

$$6 - 4 + 2 + 5i + 3i - 7i = 4 + i$$

Example 3: Find $\text{Re} [(3 + 5i) - (5 - 4i) - (-2 - 3i)]$

$$\text{Solution: } 3 + 5i - 5 + 4i + 2 + 3i$$

$$3 - 5 + 2 + 5i + 4i + 3i = 12i$$

$$\text{Re} [(3 + 5i) - (5 - 4i) - (-2 - 3i)] = 0$$

Example 4: Solve $(3 + 2i)(4 - 5i)$

$$\text{Solution: } 3(4 - 5i) + 2i(4 - 5i)$$

$$12 - 15i + 8i - 10i^2 \quad (\text{Note } i^2 = -1)$$

$$12 + 10 - 15i + 8i$$

Example 5: Evaluate: $\text{Im} [(2 + i)(3 - 2i)(-4 - 3i)]$

$$\text{Solution: } (2 + i)(3 - 2i) = 6 - 4i + 3i - 2i^2 = 6 - 2i + 2 = 4 - 2i$$

$$(4 - 2i)(-4 - 3i) = -16 - 12i + 8i + 6i^2 = -16 - 4i - 6 = -22 - 4i$$

$$\text{Im} [(2 + i)(3 - 2i)(-4 - 3i)] = -4$$

7.2 CONJUGATE COMPLEX NUMBERS

A pair of complex numbers are called conjugate complex

numbers and the product entirely real if the complex numbers are of the form $(a + bi)(a - bi)$.

The two complex numbers look the same except that the sign is different. While

$(4 + 5i)$ and $(4 - 5i)$ are conjugate complex numbers

$(5 - 3i)$ and $(5 + 3i)$ are conjugate complex numbers

$(6 + 2i)$ and $(2 + 6i)$ and $(5 - 3i)$ and $(-5 + 3i)$ are not conjugate complex numbers.

7.3 DIVISION OF COMPLEX NUMBERS

To divide one complex number by another, one must multiply both the numerator and denominator by the conjugate of the denominator. This will reduce the denominator into a real number hence, the final step can then be completed.

Example 6: Solve: $\frac{4 - 5i}{1 - 2i}$

$$\text{Solution: } \frac{4 - 5i}{1 - 2i} \times \frac{1 + 2i}{1 + 2i} = \frac{4 + 8i - 5i - 10i^2}{1 + 2i - 2i - 4i^2} = \frac{4 + 10 + 3i}{1 + 4}$$

$$= \frac{14 + 3i}{5} = \frac{14}{5} + \frac{3i}{5}$$

Example 7: Solve $\frac{(2 + 3i)(1 - 2i)}{3 + 4i}$

$$\text{Solution: } \frac{(2 + 3i)(1 - 2i)}{3 + 4i} = \frac{2 - 4i + 3i - 6i^2}{3 + 4i} = \frac{8 - i}{3 + 4i}$$

$$= \frac{8 - i}{3 + 4i} \times \frac{3 - 4i}{3 - 4i} = \frac{24 - 32i + 3i - 4i^2}{9 - 12i + 12i - 16i^2} = \frac{20 - 35i}{25}$$

$$= \frac{20}{25} - \frac{35}{25}i$$

7.4 EQUAL COMPLEX NUMBERS

Two complex number $Z_1 = a+ib$ and $Z_2 = c + id$ are said to be equal if $a=c$ and $b=d$

Example 8: If $(a+b) + i(a-b) = 7+2i$, find the values of a and b

Solution: Since $(a + b) + i(a-b) = 7 + 2i$

$$\text{Then } a + b = 7$$

$$a - b = 2$$

$$2a = 9$$

$$\therefore a = 4.5$$

$$\text{and } 4.5 + b = 7 \rightarrow b = 7 - 4.5 = 2.5$$

Example 9: If $(2 + 3i)(3-4i) = x + iy$, evaluate x and y

$$\text{Solution: } 6 - 8i + 9i - 12i^2 = x + iy$$

$$18 + i = x + iy$$

$$\therefore x = 18 \text{ and } y = 1$$

7.5 POLAR FORM OF A COMPLEX NUMBER

It is convenient sometimes to express a complex number $a + ib$ in a different form on the Argand diagram, let OP be a complex number $a + ib$.

Let r = length of the complex number and θ the angle made with OX .

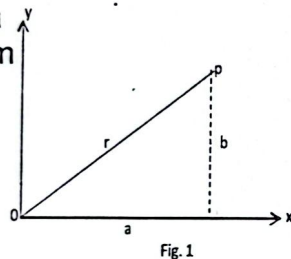
$$\text{Then } r^2 = a^2 + b^2$$

$$r = \sqrt{a^2 + b^2}$$

and

$$\tan \theta = \frac{b}{a}$$

$$\theta = \tan^{-1}\left(\frac{b}{a}\right)$$



Also, $a = r \cos \theta$ and $b = r \sin \theta$

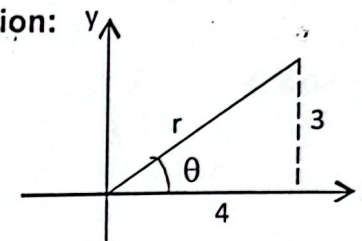
Since $z = a + ib$

$$\text{Then } z = r \cos \theta + ir \sin \theta = r(\cos \theta + i \sin \theta)$$

This is called the polar form of the complex number $a+ib$, where the modulus of z written as $|z| = \sqrt{a^2+b^2}$ and r the radius is $\sqrt{a^2+b^2}$ and the angle $\theta = \tan^{-1}(b/a)$

Example 10: Express $z = 4 + 3i$ in polar form

Solution:



$$r = \sqrt{a^2 + b^2} \rightarrow \sqrt{4^2 + 3^2} = 5$$

$$\tan \theta = \frac{3}{4} = 0.75$$

$$\theta = \tan^{-1}\left(\frac{3}{4}\right) = 36^\circ 52'$$

$$\text{Then } z = a + ib = r[\cos \theta + i \sin \theta]$$

$$= 5 [\cos 36^\circ 52' + i \sin 36^\circ 52']$$

Example 11: Calculate $|z|$ if $z = \frac{z_1 \cdot z_2}{z_3}$ Given that

$$z_1 = 2 + 3i, z_2 = 1 - 2i \text{ and } z_3 = 3 + 4i$$

$$\text{Solution: } \frac{(2 + 3i)(1 - 2i)}{3 + 4i} = \frac{8 - i}{3 + 4i} = \frac{20}{25} - \frac{35i}{25}$$

$$|z| = \sqrt{\left(\frac{20}{25}\right)^2 + \left(\frac{-35}{25}\right)^2} = \sqrt{\frac{400}{625} + \frac{1225}{625}} = \sqrt{\frac{1625}{625}}$$

$$|z| = \frac{40.31}{25} = 1.612 \text{ to 3dp}$$

7.6 EXPONENTIAL OR EULER'S FORM OF A COMPLEX NUMBER

There is still another way of expressing a complex number which we must stress due to its usefulness. It must be recalled that many functions can be expressed as series as:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$$

If we now have to replace x by $i\theta$ in the e^x Series we obtain:

$$e^{i\theta} = 1 + i + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \dots$$

$$= 1 + i\theta + \frac{i^2\theta^2}{2!} + \frac{i^3\theta^3}{3!} + \frac{i^4\theta^4}{4!} + \frac{i^5\theta^5}{5!} + \dots$$

$$= 1 + i\theta - \frac{\theta^2}{2!} + \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} - \dots$$

$$= \left[1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots \right] + i \left[\frac{\theta}{3!} - \frac{\theta^3}{5!} + \dots \right]$$

$$e^{i\theta} = \cos \theta + i \sin \theta$$

This is called the exponential form of the complex number. It can be readily obtained from the polar form, $Z = r(\cos \theta +$

$i \sin \theta)$

It is important to note, however, that in the exponential form, the angle must be in radians.

Example 12: Express $5(\cos 60^\circ + i \sin 60^\circ)$ in Euler's form

Solution: From $5(\cos 60^\circ + i \sin 60^\circ)$ $r = 5$ and $\theta = 60^\circ = \pi/3$ rad
 $\therefore$ Exponential form of $5(\cos 60^\circ + i \sin 60^\circ) = 5e^{i\pi/3}$

7.7 DEMOIVRE'S THEOREM

Suppose $Z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $Z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$

Prove that (i) $Z_1 Z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$

$$(ii) \quad \frac{Z_1}{Z_2} = \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

Solution: (1) $Z_1 Z_2 = [r_1(\cos \theta_1 + i \sin \theta_1)] [r_2(\cos \theta_2 + i \sin \theta_2)]$
 $= r_1 r_2 [\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2 + i(\sin \theta_1 \cos \theta_2 + \cos \theta_1 \sin \theta_2)]$
 $= r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$

$$(ii) \quad \frac{Z_1}{Z_2} = \frac{r_1(\cos \theta_1 + i \sin \theta_1)}{r_2(\cos \theta_2 + i \sin \theta_2)} \cdot \frac{(\cos \theta_2 - i \sin \theta_2)}{(\cos \theta_2 - i \sin \theta_2)}$$

$$\frac{Z_1}{Z_2} = \frac{r_1}{r_2} \frac{[(\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2) + i(\sin \theta_1 \cos \theta_2 - \cos \theta_1 \sin \theta_2)]}{\cos^2 \theta_2 + \sin^2 \theta_2}$$

$$= \frac{r_1}{r_2} [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

Example 13: Show that (a) $\cos \theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$ (b) $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

Solution :

(a) Recall that

$$e^{i\theta} = \cos \theta + i \sin \theta \text{ and}$$

$$e^{-i\theta} = \cos \theta - i \sin \theta$$

On adding $e^{i\theta}$ and $e^{-i\theta}$
we obtain

$$\begin{aligned} e^{i\theta} + e^{-i\theta} &= \cos \theta + i \sin \theta + \cos \theta - i \sin \theta \\ &= 2 \cos \theta \\ \therefore \cos \theta &= \frac{e^{i\theta} + e^{-i\theta}}{2} \end{aligned}$$

(b) On subtracting $e^{-i\theta}$ from $e^{i\theta}$ we obtain

$$\begin{aligned} e^{i\theta} - e^{-i\theta} &= \cos \theta + i \sin \theta - (\cos \theta - i \sin \theta) = 2i \sin \theta \\ \therefore \sin \theta &= \frac{e^{i\theta} - e^{-i\theta}}{2i} \end{aligned}$$

Exercises

- Perform each of the Indicated operations
 (a) $(4 - 3i) + (2i - 8)$ (b) $(i - 2)(1 + i) - 3(i - 1)$
 (c) $\operatorname{Re}[3(-1 + 4i) - 2(7 - i)]$ (d) $\frac{(2 + i)(3 - 2i)(1 + 2i)}{(1 - i)^2}$
- Suppose $z_1 = 1 - i$, $z_2 = -2 + 4i$ and $z_3 = 3 - 2i$. Evaluate the following:
 (a) $z_1^2 - 2z_1 + 3$
 (b) $(z_2 + z_3)(z_1 \cdot z_3)$
 (c) $|2z_2 + 3z_1|^2$
 (d) $\operatorname{Im}\{2z_1^3 - 4z_2^2 + 5z_3^2\}$

- Perform the indicated operations
 (a) $\frac{1}{2}(4 + 3i) - \frac{3}{2}(5 + 2i)$ (b) $3(1 + i) + 2(4 - 3i) - (2 + 5i)$
 (c) $|(4 - 3i) + (-1 + 2i)|$ (d) $(2i - 1)^2 \left(\frac{4}{1 - i} + \frac{2 - i}{1 + i} \right)$
- Express the following in polar form
 (a) $2 - 2i$ (b) $2\sqrt{2} + 2\sqrt{2}i$ (c) -5
- Prove that (a) $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$
 (b) $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$
- Evaluate:
 (a) $5(\cos 20^\circ + i \sin 20^\circ) \times 3(\cos 40^\circ + i \sin 40^\circ)$
 (b) $\frac{[8(\cos 40^\circ + i \sin 40^\circ)]^3}{[2(\cos 60^\circ + i \sin 60^\circ)]^4}$
- Given that $Z_1 = 2 + 5i$ and $Z_2 = 3 - i$, evaluate
 (a) $Z_1 \cdot Z_2$ (b) $|Z_2 \cdot Z_1|$
- Find the area of a triangle having vertices at $-4 - i$, $1 + 2i$ and $4 - 3i$
 (b) Find the area of a quadrilateral having vertices at $(2, -1)$, $(4, 3)$, $(-1, 2)$ and $(-3, -2)$.

CHAPTER EIGHT

8.1

TRIGONOMETRY

Angles: The concept of angles can be viewed as when a straight line is rotated about a fixed point, it sweeps out an angles that can be measured either in degrees or in radians. The rotation of a straight line through a full angle and returning to its starting position is said to have rotated through 360 degrees (360°).

A degree can be subdivided into 60 minute ($60'$) while a minute can be subdivided into 60 seconds ($60''$). While angle on a straight line is 180° , a right angle is 90° .

An angle that is measured in degrees minutes and seconds can be converted to a decimal degree vice visa as follows:

Example 1: Convert $57^\circ 36' 18''$ to decimal degree.

$$\begin{aligned}\text{Solution: } 57^\circ 36' 18'' &= 57^\circ + \frac{36}{60} + \frac{18}{60 \times 60} \\ &= (57 + 0.6 + 0.005)^\circ = 57.605^\circ \text{ to 3dp}\end{aligned}$$

Example 2: Convert $135^\circ 29' 7''$ to decimal degree

$$\begin{aligned}\text{Solution: } 135^\circ 29' 7'' &= 135^\circ + \left(\frac{29}{60}\right)^\circ + \left(\frac{7}{60 \times 60}\right)^\circ \\ &= (135 + 0.483 + 0.00194)^\circ = 135.485^\circ \text{ to 3dp.}\end{aligned}$$

Example 3: Convert 63.478° to degree minutes and seconds

$$\begin{aligned}\text{Solution: } 63.478^\circ &= 63^\circ + (0.478 \times 60)' \\ &= 63^\circ + 28.68' \\ &= 63^\circ + 28' + (0.68 \times 60)'' \\ &= 63^\circ + 28' + 40.8'' = 63^\circ 28' 41'' \text{ to} \\ &\text{meant second}\end{aligned}$$

Example 4: Convert 236.986° to degree, minutes and seconds

$$\begin{aligned}\text{Solution: } 236.986^\circ &= 236^\circ + (0.986 \times 60)' \\ &= 236^\circ + 59.16' \\ &= 236^\circ + 59' + (0.16 \times 60)'' \\ &= 236^\circ + 59' + 9.6'' = 236^\circ 59' 10''\end{aligned}$$

Degree Radians: An alternative unit of measure of an angle is the radian

$$360^\circ = 2\pi \text{ rad} = 6.2831 \text{ rad}$$

$$1^\circ = \frac{\pi}{180} \text{ rad} = 0.0175 \text{ rad to 3 sig. fig.}$$

after when degree are changed to radians, they are better within as multiples of π as:

$$360^\circ = 2\pi \text{ rad}, 180^\circ = \pi \text{ rad}, 90^\circ = \pi/2 \text{ rad}, 45^\circ = \pi/4 \text{ rad e.t.c}$$

Example 5: Convert 120° to radian

$$\begin{aligned}\text{Solution: } 360^\circ &= 2\pi \text{ rad} \\ 120^\circ &= x \\ \therefore x &= \frac{120^\circ \times 2\pi}{360^\circ} \text{ rad} = \frac{2\pi}{3} \text{ rad}\end{aligned}$$

Example 6: Change 270° to radian

$$\begin{aligned}\text{Solution: } 360^\circ &= 2\pi \text{ rad} \\ 270^\circ &= y\end{aligned}$$

$$\therefore y = \frac{270^\circ \times 2 \text{ rad } \pi}{360^\circ} = \frac{3\pi}{2} \text{ rad}$$

Example 7: Express 2.34 rad in degrees

Solution: $2.34 \text{ rad} = \frac{2.34 \times 180^\circ}{\pi} = 134.1^\circ \text{ to 1 d.p.}$

Example 8: Express $\frac{7\pi}{4}$ rad in degrees

Solution: $\frac{7\pi}{4} \text{ rad} = \frac{7\pi}{4} \times \frac{180^\circ}{\pi} = 315^\circ$

8.2 TRIGONOMETRIC RATIOS

Given a right-angled triangle ABC as in fig 1
With angle θ at vertex B where AC is opposite
Side BC is adjacent to θ and side AB is called
the hypotenuse

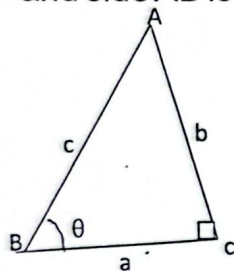


Fig. 1

Trigonometric ratios are defined herein as:

Sine of angle = $\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{AC}{AB}$

Cosine of angle = $\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}} = \frac{BC}{AB}$

Tangent of angle = $\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}} = \frac{AC}{BC} = \frac{\sin \theta}{\cos \theta}$

8.3 RECIPROCAL RATIOS

In addition to the three fundamental trigonometrical ratios, there are reciprocal ratios namely

Cosec $\theta = \frac{1}{\sin \theta}$

Sec $\theta = \frac{1}{\cos \theta}$

Cot $\theta = \frac{1}{\tan \theta} = \frac{\cos \theta}{\sin \theta}$

All right-angled triangles have a property in common that is expressed in Pythagoras' theorem such as:

The Square on the hypotenuse of a right-angled triangle is equal to the sum of the squares on the other two sides.
i.e.

$$a^2 + b^2 = c^2$$

8.4 SPECIAL TRIANGLES

Two right-angled triangles are of special interest here because the trigonometric ratios of their angles can be written in surd or fractional form

Isosceles right-angled triangle with two sides and two angles equal as 90° , 45° and 45° with the lengths in the ratio 1:1:2 by Pythagoras' theorem

$$\sin 45^\circ = \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\tan 45^\circ = 1$$

OR

$$\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{1}{\sqrt{2}}$$

$$\text{and } \tan \frac{\pi}{4} = 1$$

Half Equilateral Triangles

The second right-angled triangle of interest is the half equilateral triangle with side

lengths in the ratio $1:\sqrt{3}:2$

$$\sin 30^\circ = \cos 60^\circ = \frac{1}{2}$$

$$\sin 60^\circ = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

$$\tan 60^\circ = \frac{1}{\tan 30^\circ} = \sqrt{3}$$

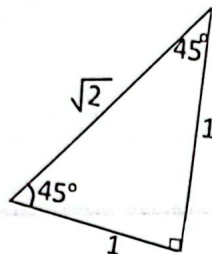


Fig. 2

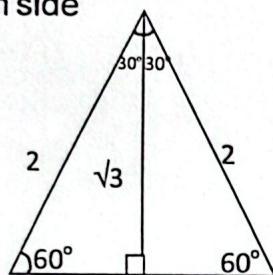


Fig 3.

8.5 TRIGONOMETRIC IDENTITIES

Given the right angles triangle as in fig 1 with vertices A, B and C side opposite the vertices of a, b and hypotenuse C and angle θ at B

Then: $a^2 + b^2 = C^2$ [By Pythagoras' theorem]

$$\sin^2 \theta + \cos^2 \theta = 1$$

This is called the fundamental trigonometrical identity for which we obtain the followings:

$$\sin^2 \theta = 1 - \cos^2 \theta \rightarrow \sin \theta = \sqrt{1 - \cos^2 \theta}$$

$$\cos^2 \theta = 1 - \sin^2 \theta \rightarrow \cos \theta = \sqrt{1 - \sin^2 \theta}$$

Other identities derived from the fundamental identity include:

$$1 + \tan^2 \theta = \sec^2 \theta \rightarrow \tan^2 \theta = \sec^2 \theta - 1$$

$$1 + \cot^2 \theta = \operatorname{cosec}^2 \theta \rightarrow \cot^2 \theta = \operatorname{cosec}^2 \theta - 1$$

8.6 COMPOUND ANGLES IDENTITIES

The Trigonometric ratios of the sum or difference of two angles can be given in terms of the ratios of the individual angles. For instance, the cosine of a sum of angles is given by

$$\cos (A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos (A-B) = \cos A \cos B + \sin A \sin B$$

$$\sin (A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin (A-B) = \sin A \cos B - \cos A \sin B$$

$$\tan (A+B) = \frac{\sin (A+B)}{\cos (A+B)} = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan (A-B) = \frac{\sin (A-B)}{\cos (A-B)} = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

8.7 DOUBLE ANGLES

Double angles formulas are identities drawn from the sum and differences of angles as in compound angles when $A=B$ as

$$\sin 2A = \sin (A+A) = \sin A \cos A + \cos A \sin A = 2 \sin A \cos A$$

$$\begin{aligned} \cos 2A &= \cos (A+A) = \cos A \cos A - \sin A \sin A \\ &= \cos^2 A - \sin^2 A = \cos^2 A - (1 - \cos^2 A) \\ &= 2 \cos^2 A - 1 \end{aligned}$$

$$\tan 2A = \tan (A+A) = \frac{\tan A + \tan A}{1 - \tan A \tan A} = \frac{2 \tan A}{1 - \tan^2 A}$$

SUMS AND DIFFERENCES OF RATIOS

$$\sin A + \sin B = 2 \sin \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

$$\sin A - \sin B = 2 \cos \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

$$\cos A + \cos B = 2 \cos \left(\frac{A+B}{2} \right) \cos \left(\frac{A-B}{2} \right)$$

$$\cos A - \cos B = 2 \sin \left(\frac{A+B}{2} \right) \sin \left(\frac{A-B}{2} \right)$$

8.8 PRODUCTS OF RATIOS

$$2 \sin A \cos B = \sin (A+B) + \sin (A-B)$$

$$2 \cos A \cos B = \cos (A+B) + \cos (A-B)$$

$$2 \sin A \sin B = \cos (A-B) - \cos (A+B)$$

Example 9: Show that $\tan \theta + \cot \theta = \sec \theta \operatorname{cosec} \theta$

$$\begin{aligned} \text{Solution: LHS} \rightarrow \tan \theta + \cot \theta &= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} \\ &= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta} = \frac{1}{\sin \theta \cos \theta} \\ &= \frac{1}{\sin \theta} \cdot \frac{1}{\cos \theta} = \sec \theta \operatorname{cosec} \theta \text{ RHS.} \end{aligned}$$

Example 10: Prove that $\tan^2 \theta - \sin^2 \theta = \sin^4 \theta \sec^2 \theta$

$$\begin{aligned} \text{Solution: LHS} \rightarrow \tan^2 \theta - \sin^2 \theta &= \frac{\sin^2 \theta}{\cos^2 \theta} - \frac{\sin^2 \theta}{1} \\ &= \frac{\sin^2 \theta}{1} \cdot \frac{1}{\cos^2 \theta} - \sin^2 \theta \\ &= \sin^2 \theta \sec^2 \theta - \sin^2 \theta \\ &= \sin^2 \theta [\sec^2 \theta - 1] = \sin^2 \theta \tan^2 \theta \\ &= \sin^2 \theta \cdot \frac{\sin^2 \theta}{\cos^2 \theta} = \sin^4 \theta \cdot \frac{1}{\cos^2 \theta} \\ &= \sin^4 \theta \sec^2 \theta \text{ RHS} \end{aligned}$$

Example 11: Evaluate $\cos 75^\circ$ without using table

Solution: $\cos 75^\circ = \cos (30^\circ + 45^\circ) = \cos 30^\circ \cos 45^\circ - \sin 30^\circ \sin 45^\circ$

$$\begin{aligned} &= \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \\ &= \frac{\sqrt{3}}{2} - \frac{1}{2\sqrt{2}} = \frac{\sqrt{3}-1}{2\sqrt{2}} \times \frac{2\sqrt{2}}{2\sqrt{2}} \\ &= \frac{2\sqrt{6}-2\sqrt{2}}{8} = \frac{1\sqrt{6}}{4} - \frac{1\sqrt{2}}{4} = \frac{1(\sqrt{6}-\sqrt{2})}{4} \end{aligned}$$

Example: Show that $\sin 75^\circ = \frac{1+\sqrt{3}}{2\sqrt{2}}$

Solution: $\sin 75^\circ = \sin (45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$

$$\begin{aligned} &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\ &= \frac{\sqrt{3} + 1}{2\sqrt{2}} = \frac{\sqrt{3} + 1}{2\sqrt{2}} \end{aligned}$$

Example 13: Show that $(\sin \theta - \cos \theta)^2 + (\sin \theta + \cos \theta)^2 = 2$

Solution: LHS $(\sin \theta - \cos \theta)(\sin \theta - \cos \theta) + [(\sin \theta + \cos \theta)(\sin \theta + \cos \theta)]$
 $(\sin^2 \theta + \cos^2 \theta) + (\sin^2 \theta + \cos^2 \theta) = 1 + 1 = 2 \text{ RHS.}$

Refer

1.

TUTORIAL QUESTIONS

1a. Convert the angle $68^\circ 47' 2''$ to decimal degree form

2.

(b) Express 65.308° in degree, minutes and seconds

(c) Convert 69.12° radians correct to 2 d.p.

(d) Convert the radians 2.139 to degrees to 2 d.p.

2. Verify each of the following trigonometric identities

3.

(a) $\frac{(\cos \alpha - \sin \alpha)^2}{\cos \alpha} = \sec \alpha - 2 \sin \alpha$

4.

(a) $\frac{\operatorname{cosec} \alpha \sec \alpha}{\cot \alpha} = 1 + \tan^2 \alpha$

3. Show that (a) $\tan 15^\circ = \frac{\sqrt{3}-1}{\sqrt{3}+1}$ (b) $\sin 75^\circ = \frac{1+\sqrt{3}}{2\sqrt{2}}$

5.

(4) Verify (a) $\frac{\cos \theta - 1}{\sec \theta + \tan \theta} + \frac{\cos \theta + 1}{\sec \theta - \tan \theta} = 2(1 + \tan \theta)$

6

(5) Show that $\cot \theta \cos \theta + \tan \theta \sin \theta = (\operatorname{cosec} \theta + \sec \theta)(1 - \sin \theta \cos \theta)$

7.